

Big Data Begins to Crack the Case of Endometriosis

Records from millions of patients at University of California health centers found correlations between endometriosis, one of the most common diseases in women, and a bounty of other diseases.

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Scientists at UC San Francisco have found that endometriosis — a painful chronic disease that often goes undiagnosed yet is estimated to affect as many as 200 million women worldwide — frequently occurs alongside conditions like cancer, Crohn's disease, and migraine.

The research could improve diagnosis and, ultimately, treatments for endometriosis, preventing women from having to go on long diagnostic journeys in which they are told that nothing is wrong with them.

The study, which appeared in [Cell Reports Medicine](#) on July 31, used computational methods developed at UCSF to analyze anonymized patient records collected at the University of California's six health centers.

"We now have both the tools and the data to make a difference for the huge population that suffers from endometriosis," said Marina Sirota, PhD, the interim director of the UCSF Bakar Computational Health Sciences Institute (BCHSI), professor of pediatrics, and senior author of the paper. "We hope this can spur a sea change in how we approach this disorder."

"The impact on patients' lives is huge"

Endometriosis, often called 'endo,' occurs when the endometrium, the blood-rich tissue that grows in the uterus before being expelled each month during menstruation, spreads to other nearby organs. It causes chronic pain and infertility. It is estimated that nearly 10% of women worldwide suffer from it.

"Endo is extremely debilitating," said Linda Giudice, MD, PhD, MSc, a physician-scientist in the Department of Obstetrics, Gynecology and Reproductive Sciences at UCSF and co-author of the paper. "The impact on patients' lives is huge, from their interpersonal relationships to being able to hold a job, have a family, and maintain psychological wellbeing."

The gold standard to diagnose endometriosis is surgery to find endometrial tissue outside of the uterus, and it is mainly treated with hormones to suppress the menstrual cycle, or surgery to remove the excess tissue.

But not everyone responds to hormonal therapy, which can have debilitating side effects. Even after surgery, the condition can flare up. Removal of the uterus is a last-ditch measure that is usually reserved for older women; but some women continue to experience pain even after a hysterectomy.

Giudice partnered with Sirota to leverage [the UC health system's anonymized patient data](#) against endo, which can vary dramatically across patients. Both Giudice and Sirota are principal investigators at the UCSF-Stanford Endometriosis Center for Discovery, Innovation, Training and Community Engagement ([ENACT](#)).

"This data is messy; it was not collected for research purposes but for the real, human purpose of helping women who need care," Sirota said. "We had the rare chance to rigorously assess how endometriosis presents across UCSF's patient population and then ask whether these observations held true with patients seen at the other UC health centers."

Data connects the dots for understanding endometriosis

Using algorithms developed for the task, Umair Khan, a bioinformatics graduate student in Sirota's lab and first author of the paper, hunted for connections linking endometriosis with the rest of each patient's health history.

He compared endo patients with patients who did not have it, and categorized the patients with endo into groups based on shared health histories. He mapped his findings from the UCSF data against the rest of the UC's health data to see if they held up across California.

"We found over 600 correlations between endometriosis and other conditions," Khan said. "These ranged from what we already knew or suspected, like infertility, autoimmune disease, and acid-reflux, to the unexpected, like certain cancers, asthma, and eye-related diseases."

Some patients had migraines, bolstering previous studies suggesting that migraine drugs might help treat endometriosis.

"In the past, studies like this would have been nearly impossible," said Tomiko Oskotsky, MD, an investigator at ENACT, associate professor in UCSF BCHSI, and co-author of the paper. "It was only 12 years ago that de-identified electronic health records became available at this scale."

The study supports the growing understanding of endometriosis as a "multi-system" disorder — a disease arising from dysfunction throughout the body.

“This is the kind of data we need to move the needle, which hasn’t moved in decades,” Giudice said. “We’re finally getting closer to faster diagnosis and, eventually, we hope, tailored treatment for the millions of women who suffer from endometriosis.”

Authors: Other UCSF authors are Bahar D. Yilmaz, MD, Jacquelyn Roger, Ketrin Gjoni, and Juan C. Irwin, MD, PhD. For all authors, see the paper.